

Amartya Tashi Mitra

Curriculum Vitae

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[website](#)

Areas of Interest

Evolution, Development, Morphology, Vision

Education

- 2022 - Current, **University of Cincinnati, USA**
PhD in Biology supervised by Professor Elke K. Buschbeck
Thesis - Towards understanding the development of stemmata
- June - July 2025, **Marine Biological Laboratory, USA**
Embryology: Concepts and Techniques in Modern Developmental Biology
- 2019 - 20, **University College London, UK**
MRes in Biodiversity Evolution and Conservation. **Overall mark: Distinction (75%)**
 - Masters research project at the **Natural History Museum, London**, supervised by Dr Kate Thomas, Dr David Gower and Dr Jeffrey Streicher on "Lens morphology is influenced by ecology in frogs and toads (Amphibia: Anura)", **Distinction (80%)**
 - Masters research project at the **Zoological Society of London**, supervised by Dr Gonçalo Rosa, Professor Trenton Garner on "Disentangling the impacts of different stressors in a complex host pathogen system", **Distinction (74%)**
- 2016 - 19, **Mumbai University, St. Xavier's College, India**
Integrated BSc, majoring in Zoology and Biochemistry, **Distinction (78%)**
 - Courses - Evolutionary biology, Ethology, Entomology, Biostatistics, Biochemical Pathways
- 2015, **Delhi Public School, Navi Mumbai, India**
 - Subjects - Biology (96%), Physics (95%), Chemistry (92%), Psychology (83%), English (91%)

Professional Appointments

- 2023 - 26, **Biology Graduate Student Association, University of Cincinnati, USA**
Faculty Representative
- 2022, **Biology Graduate Student Association, University of Cincinnati, USA**
Graduate Student Government Representative
- June - August 2019, **Bombay Natural History Society, India**
Intern - Wetlands Programme. Organisation and digitisation of long-term bird ringing data (Supervisor: Tuhina Katti)
- September 2018 - September 19, **Nature Conservation Foundation, Mumbai, India**
Part-time volunteer. Collected data on shark landing and bycatch from fisheries for conservation planning
- April - May 2018, **Universidad San Francisco de Quito, Ecuador.**
Volunteer - DNA extraction and PCR, bird captures using mist nets (Supervisor: Dr Elisa Bonaccorso)
- December 2018, 2017, **Green Line Environmental NGO, Mumbai, India**
Volunteer - Conducting bird censuses for a biodiversity assay of Aarey, Mumbai, India

- 2018, **Zoology board of Studies, Mumbai University, St Xavier's College, India.**
Student member - suggesting revisions to the second year BSc Zoology syllabus

Publications

- Mitra A.T.**, Rathore S., Jester A., Hyland-Brown R., Hassert J., Benoit J.B., Stowasser A., Buschbeck E.K. (2025) Cataract induction in an arthropod reveals how lens crystallins contribute to the formation of biological glass. *PLoS One* 20.6 e0325229. DOI: [10.1371/journal.pone.0325229](https://doi.org/10.1371/journal.pone.0325229)
- Rathore S., **Mitra A.T.**, Hyland-Brown R., Jester A., Layne J.E., Benoit J.B., Buschbeck E.K. (2024) Osmosis as nature's method for establishing optical alignment. *Curr. biol* 34.7: 1569-1575. DOI: [10.1016/j.cub.2024.02.052](https://doi.org/10.1016/j.cub.2024.02.052)
- Mitra A.T.**, Womack M.C., Gower D.J., Clark B., Streicher J.W., Bell R.C., Schott R.K., Fujita M.K., Thomas K.N. (2022) Ocular lens morphology is influenced by ecology in frogs and toads. *Proc. R. Soc. B.* 289: 20220767. DOI: [10.1098/rspb.2022.0767](https://doi.org/10.1098/rspb.2022.0767)
- Rosa G.M., Botto G.A., **Mitra A.T.**, de Almeida J.S., Hofmann M., Leung W.T., Alves de Matos A.P., Caeiro M.F., Froufe E., Loureiro A., Price S.J., Owen C., Rebelo R., Soares C. (2022). Invasive fish disrupt host-pathogen dynamics leading to amphibian declines. *Biol. Conserv.* 276, 109785. DOI: [10.1016/j.biocon.2022.109785](https://doi.org/10.1016/j.biocon.2022.109785)

Awards and Fellowships

- 2024 - Present, **Graduate Assistant Scholarship, University Graduate Assistantship**, University of Cincinnati (\$27,775 annually)
- 2026 - **Research Discovery & Endowment Fund**, University of Cincinnati (\$500)
- 2025 - **Hibbitt Endowed Education Fund**, Marine Biological Laboratory (\$7,200)
- 2024, **Graduate Student Government Graduate Research Fellowship Award**, University of Cincinnati (\$1,700)
- 2023 - 24 - **IRiS Graduate Research Fellowship**, University of Cincinnati (\$3,500)
- 2023 - **Carl Gans Travelling Fellowship**, Society for the Study of Amphibians and Reptiles (\$800)
- 2022 - 26 - **Weiman/Benedict Award**, University of Cincinnati (\$3,166)
- 2019 - 20, **Ravi Sankaran Fellowship**, Inlaks Shivdasani Foundation (\$53,366)

Conferences and Presentations

- June 2026, From compound to camera: developmental repatterning of larval eyes in insects. **Novelty, Co-option & Divergence During Gene Network Evolution** Workshop, East Sussex, UK. (Talk)
- October 2025, How lens crystallins form biological glass, **Institute for Research in Sensing (IRiS) Ignite** Biennial Meeting. (Talk)
- September 2025, Establishing molecular parallels between eye types, **Society for Developmental Biology** Midwest Regional Meeting. (Poster/Flash Talk)
- January 2025, Investigating the role of the transcription factor Cut in the development of insect camera-type eyes, **Society for Integrative and Comparative Biology (SICB) Annual Meeting**. (Talk)
- January 2024, Deconstructing a complex lens, **Society for Integrative and Comparative Biology (SICB) Annual Meeting**. (Talk)
- July 2023, Deconstructing a complex lens, **International Conference on Invertebrate Vision**

- (ICIV) Bäckaskog Castle, Sweden (Poster)
4. July 2023, **Invited speaker**: Ocular lens morphology is influenced by ecology in frogs and toads, **Joint Meeting of Ichthyology and Herpetology (JMIH; Amphibian Vision Symposium)**, Norfolk, Virginia, USA. (Talk)
 5. November 2022, Exploring the architecture of a complex lens, **Sensorium**, (Regional conference on sensory biology), Cincinnati, Ohio, USA. (Poster)
 6. January 2021, Lens morphology is influenced by ecology in frogs and toads (Amphibia: Anura) **Society for Integrative and Comparative Biology (SICB) Virtual Meeting**. (Talk)
 7. March 2020, The Evolution of Lenses in Frogs and Toads **Natural History Museum**, London, UK. (Talk)

Undergraduate Teaching Experience

1. 2024, **Undergraduate Research Mentor** to Augusta Jester, **University of Cincinnati**, USA
2. 2023 - 25, Teaching Assistant, **Neurophysiology Laboratory, University of Cincinnati**, USA
3. 2022, 24, Teaching Assistant, **Introductory Biology Laboratory, University of Cincinnati**, USA

Science Communication & Outreach

1. 2024, **volunteer** for **Biology Day**; outreach event conducting visual ecology experiments with high-school students. University of Cincinnati, Ohio, USA
2. 2023, participant in the Annual Institute of Research in Sensing (IRiS) **Light Working Group** on the importance of light for humankind and the planet. To be developed into a book (In preparation). University of Cincinnati, Ohio, USA
3. 2023, **volunteer** for **Earth Day outreach**; public engagement event on arthropod biology with an interactive live-insect booth. Rhinegeist Brewery, Cincinnati, Ohio, USA
4. 2019, wrote on brood parasites for **The Quagga**, science blog of the SciComm students of University College London, the Natural History Museum and the Zoological Society of London
5. 2019, wrote on Lepidodendrales for Volume 1, **Acta Botanica**, annual departmental magazine of the botany department. St. Xavier's College, Mumbai, India
6. 2018 **content editor, designer** and 2019 **Student editor-in-chief** for **Imprint**, annual magazine of the department of zoology. St. Xavier's College, Mumbai, India
7. 2018, wrote on raft building in fire-ants for **Lignum Vitae**, annual departmental magazine of the life sciences department. St. Xavier's College, Mumbai, India
8. 2017, **graphic designer** for **Paradigm**, the St. Xavier's College Science festival

Research Skills:

Statistics & Programming

- Foundational statistics, linear & non-linear modelling and model selection methods
- Phylogenetic comparative methods in R (MSc level course taught by Dr Natalie Cooper)
- R and R Studio for statistical computing (Course taught by Dr Natalie Cooper, NHM London)
- Git and GitHub (Course taught by Dr Natalie Cooper, NHM, London)
- Basic Python for programming
- Basic MATLAB for programming

Imaging

- Micro CT (scanning, staining and specimen preparation)
- Avizo software for 3D visualisation and analysis

- Transmission electron microscopy (imaging, staining and sample preparation)
- Scanning electron microscopy (sample preparation, imaging)
- Confocal, light sheet microscopy (imaging, sample preparation and analysis)
- Micro-ophthalmoscopy

Molecular techniques

- RNA interference in arthropods
- Quantitative PCR
- Immunohistochemistry
- *In-situ* hybridisation

Physiology

- Electroretinography (ERG)
- Intracellular electrophysiology

Insect rearing

- Coleoptera (Sunburst diving beetle, *Thermonectus marmoratus*)
- Hymenoptera (Red-headed Pine Sawfly, *Neodiprion lecontei*)
- Blattodea (Madagascar hissing cockroach, *Gromphadorhina portentosa*)
- Diptera (*Drosophila melanogaster*, *Calliphora*)
- Lepidoptera (Papilionidae, Sphingidae, Lycaenidae)

Field

- Transect sampling, mist nets bird capture for blood sampling, morphometric measurements
- Taxonomy and identification of birds (Indian subcontinent), sharks (Arabian Sea)

Population Modelling

- SCTI Tools (Vortex, Outbreak and MetaModel Manager softwares for population viability analysis, species risk assesment and metamodel analysis)

Scientific illustration

- Traditional media (watercolour, acrylic, pencil, marker, sculpture),
- Adobe, Affinity photoshop, indesign and illustrator for graphic design
- Skeleton assembly and taxidermy (course taught by Dr Santosh Gaikwad)
- Photography, videography

Languages:

Bengali, English, Hindi, Marathi, basic Spanish